Exercise 6 — Basis for Column Space and Kernel

Given a matrix,

$$A = \begin{bmatrix} 1 & 2 & 2 & 1 \\ 2 & 4 & 6 & 0 \\ 3 & 6 & 8 & 1 \end{bmatrix}$$

(a) Find a basis for the column space.
(b) Find a basis for the kernel.

Exercise 5 — Solution

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Exercise 6 — Change of basis

Let $\mathcal{B} = \{v_1, v_2\}$ with $v_1 = (1,1)^\top$, $v_2 = (3,-2)^\top$, and let $\mathcal{E}$ be the standard basis of $\mathbb{R}^2$. For a basis $\mathcal{B}$, write $[x]_\mathcal{B}$ for the *coordinate vector* of x: the unique scalars (c_1, c_2) with $x = c_1 v_1 + c_2 v_2$ (uniqueness is Proposition 8).

(a) Show $\mathcal{B}$ is a basis of $\mathbb{R}^2$ and compute $[x]_\mathcal{B}$ for $x = (5,0)^\top$.
(b) Let $P = [v_1\ v_2]$. Show that $P[x]_\mathcal{B} = x$ for every x, so P is the transition matrix from $\mathcal{B}$-coordinates to standard coordinates and P^{-1} goes the other way. Why must P be invertible?
(c) Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ have standard matrix $A = \left(\begin{smallmatrix} 0.5 & 0.3 \\ 0.2 & 0.6 \end{smallmatrix}\right)$, i.e. $[T(x)]_\mathcal{E} = A[x]_\mathcal{E}$. Show that the matrix representing T in $\mathcal{B}$-coordinates is $P^{-1}AP$, and compute it. What have you just re-derived?
(d) Now let $\mathcal{C} = \{(1,0)^\top, (1,1)^\top\}$ with $Q = [c_1\ c_2]$. Find the matrix taking $\mathcal{B}$-coordinates directly to $\mathcal{C}$-coordinates, and verify it by converting v_1 and v_2 by hand.
(e) Explain why $\operatorname{tr}(P^{-1}AP) = \operatorname{tr}(A)$ and $\det(P^{-1}AP) = \det(A)$ for *any* invertible P.

Exercise 6 — Solution

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Exercise 7 — Inner product spaces

Let

$$M = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \langle u, v \rangle_M := u^\top M v \quad \text{on } \mathbb{R}^2.$$

(a) Verify M is positive definite. Then show $\langle \cdot, \cdot \rangle_M$ satisfies the three inner product axioms, and identify exactly which property of M each axiom requires. What goes wrong if M is symmetric but only positive *semi*-definite?

(b) Compute $\|e_1\|_M$, $\|e_2\|_M$ and $\langle e_1, e_2 \rangle_M$ for the standard basis vectors. Are e_1, e_2 orthogonal in this space? Verify the Cauchy–Schwarz inequality for this pair.

(c) Apply Gram–Schmidt to $\{e_1, e_2\}$ with respect to $\langle \cdot, \cdot \rangle_M$ to produce an M-orthonormal basis $\{u_1, u_2\}$. Recall

$$w_2 = e_2 - \frac{\langle e_2, w_1 \rangle_M}{\langle w_1, w_1 \rangle_M} w_1.$$

(d) Let $R = [u_1\ u_2]$. Compute $R^\top M R$. What have you found, and how does it relate to Theorems 10 and 11 on the last slide of Bruno's notes?

Exercise 7 — Solution

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